

SOFTWARE ENGINEERING

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Scrum Project

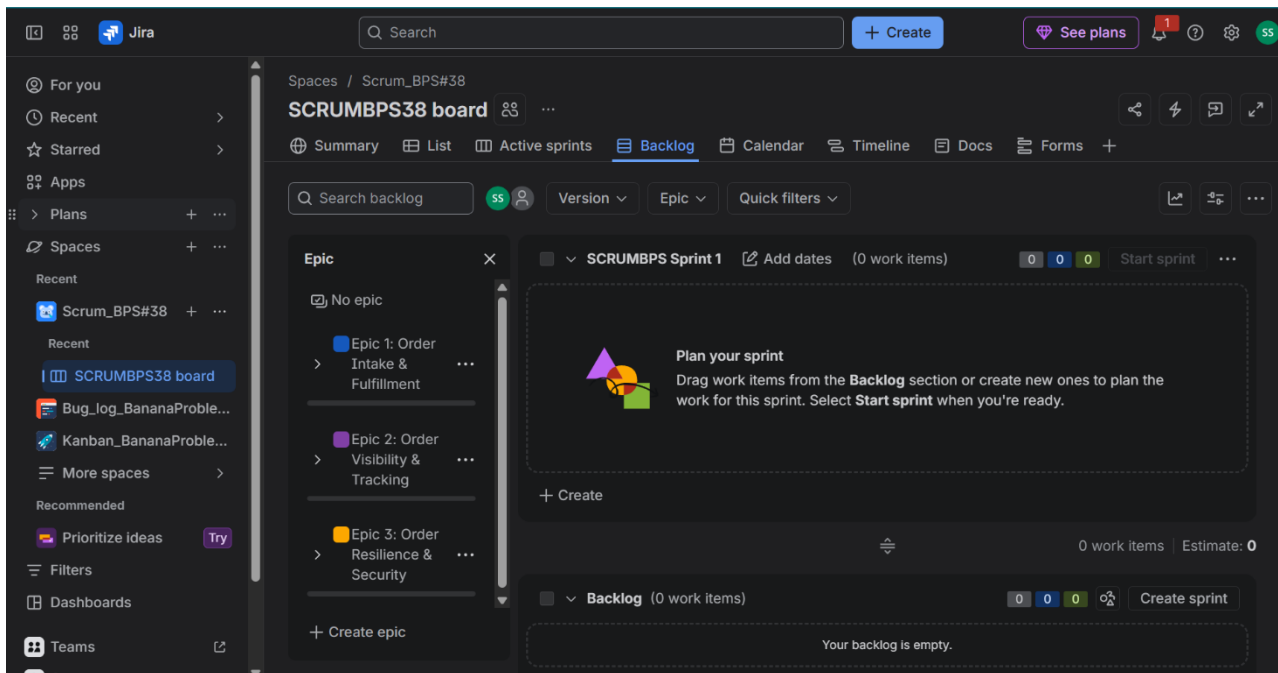
Problem Statement #38: Dropshipping Inventory & Order Orchestrator

Sriram S

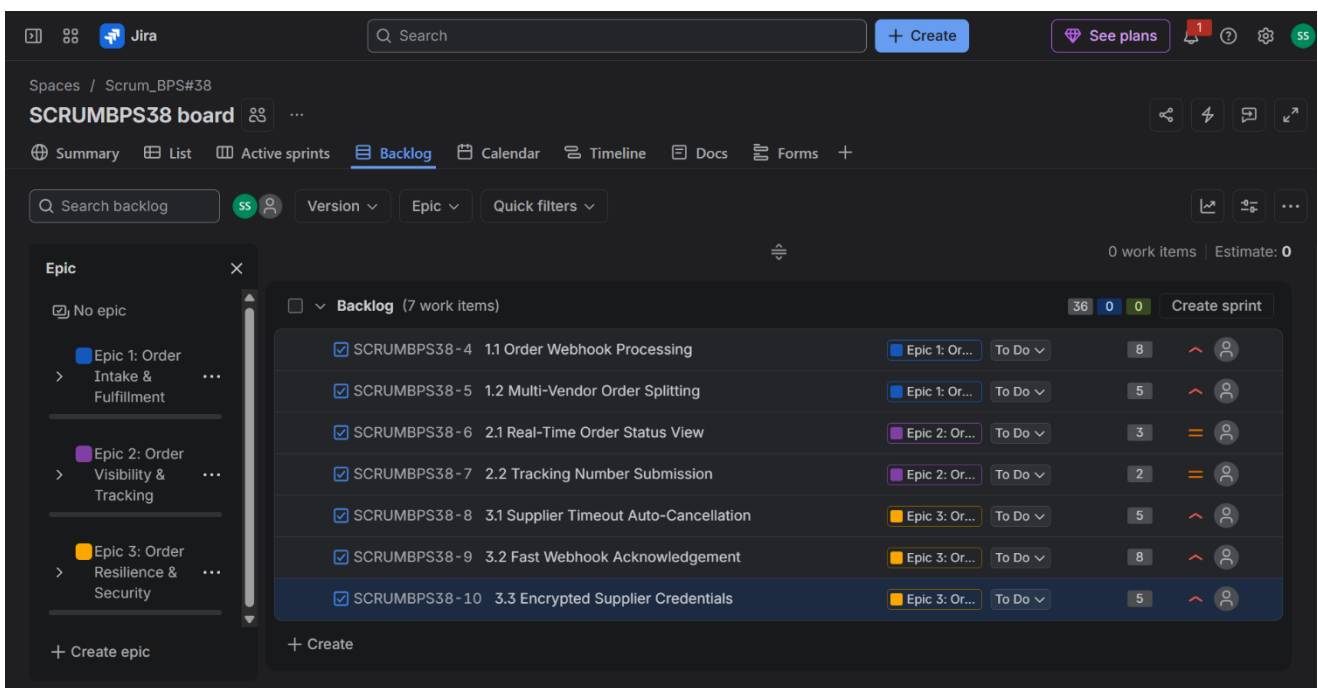
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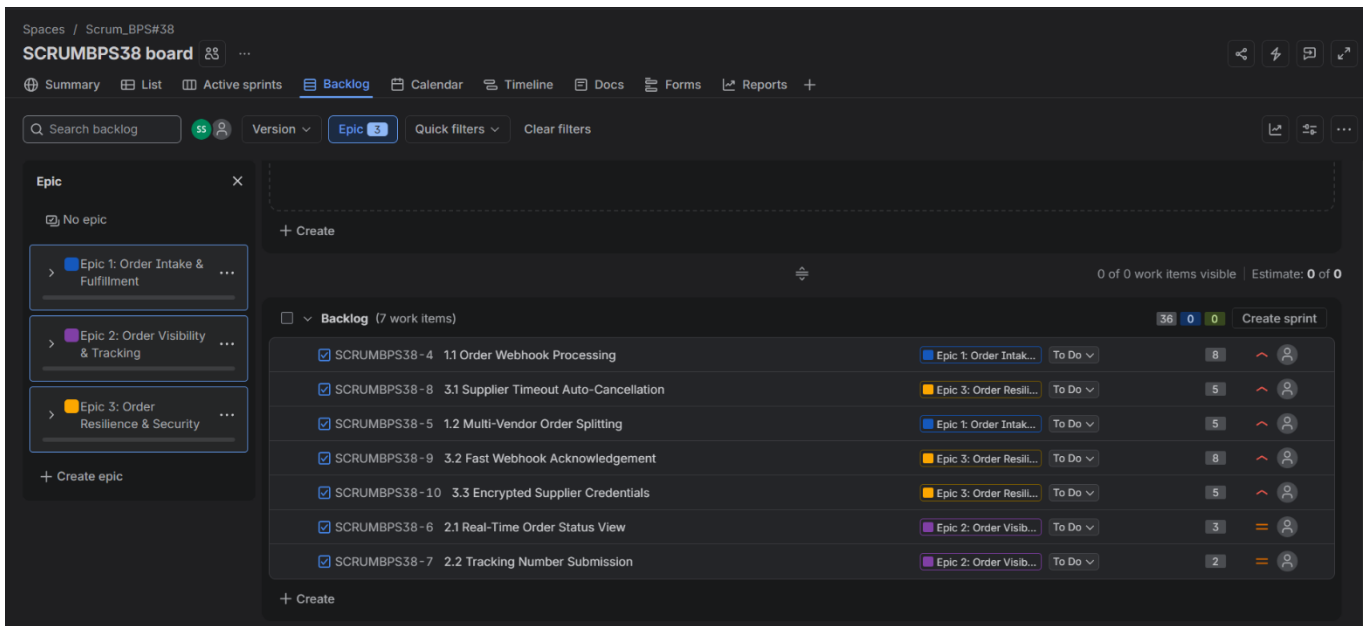
Screenshots:



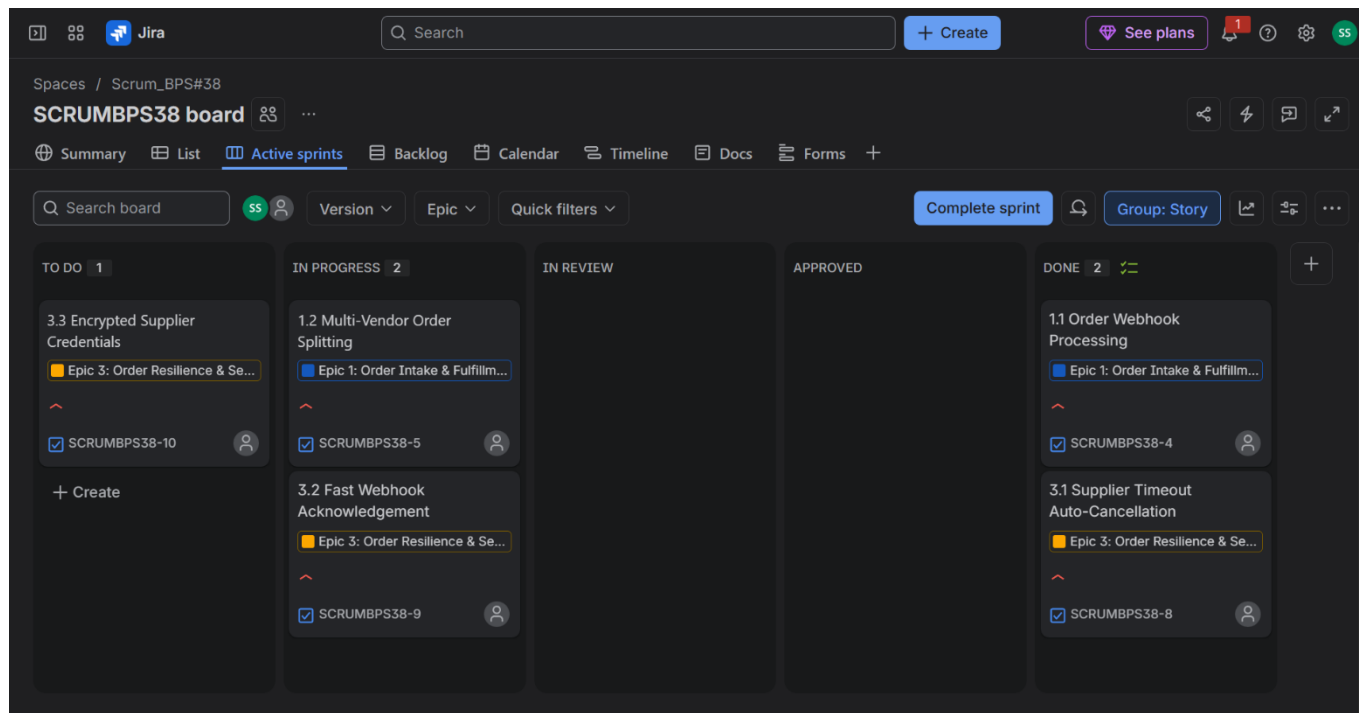
All 3 Epics created for the Dropshipping Inventory & Order Orchestrator, derived from the Lab 1 functional requirements: Order Intake & Fulfillment, Order Visibility & Tracking, and Order Resilience & Security.



7 User Stories created and mapped to their parent Epics, covering all 5 Functional Requirements and 2 Non-Functional Requirements. Backlog prioritized by business impact — High priority assigned to core order-processing, resilience, and security stories; Medium priority assigned to order-visibility features.



Story Points assigned to all 7 backlog items using the Fibonacci scale, based on estimated complexity, effort, and uncertainty.



Sprint 1 in progress on the Active Sprints board, showing stories distributed across To Do, In Progress, and Done as the sprint was simulated.



Burndown Chart for Sprint 1, showing story points completed over time against the planned guideline.

Reflection Questions:

1. Did your estimations reflect the actual effort?

Ans: The estimates were based on perceived complexity rather than actual measured effort, since this was a simulation and no real code was written. Higher-complexity stories like webhook processing and fast acknowledgement (8 points each) reflected the difficulty of real-time verification and strict performance constraints, while simpler stories like tracking number submission (2 points) involved minimal logic. In a real sprint, these estimates would likely shift once the team hit unexpected issues, such as inconsistent supplier API responses.

2. Was your backlog well-prioritized?

Ans: Yes, largely. Core functionality (order intake, multi-vendor splitting, supplier timeout handling) and non-functional concerns (performance, security) were marked High priority, since failures there would break the core pipeline or expose sensitive data. Visibility features (order status, tracking numbers) were marked Medium, since they improve experience but aren't blocking.

3. How did your simulated sprint align with your plan?

Ans: Both sprints completed exactly as planned, with every story moving cleanly from To Do to Done. This is expected in a single-person simulation with no external dependencies or blockers. A real sprint would likely require mid-sprint replanning that this simulation didn't need to account for.

4. What insights did the burndown chart give about your team's capacity?

Ans: The burndown line reaching zero by sprint end shows the committed story points (31 points in Sprint 1) were an achievable target for the one-week timebox. Since actual progress tracked close to the ideal guideline rather than lagging or finishing early, the estimate was reasonably well-calibrated. In a real team, consistently matching the guideline builds confidence in future velocity planning, while repeated early finishes or missed guidelines would signal under- or over-commitment.